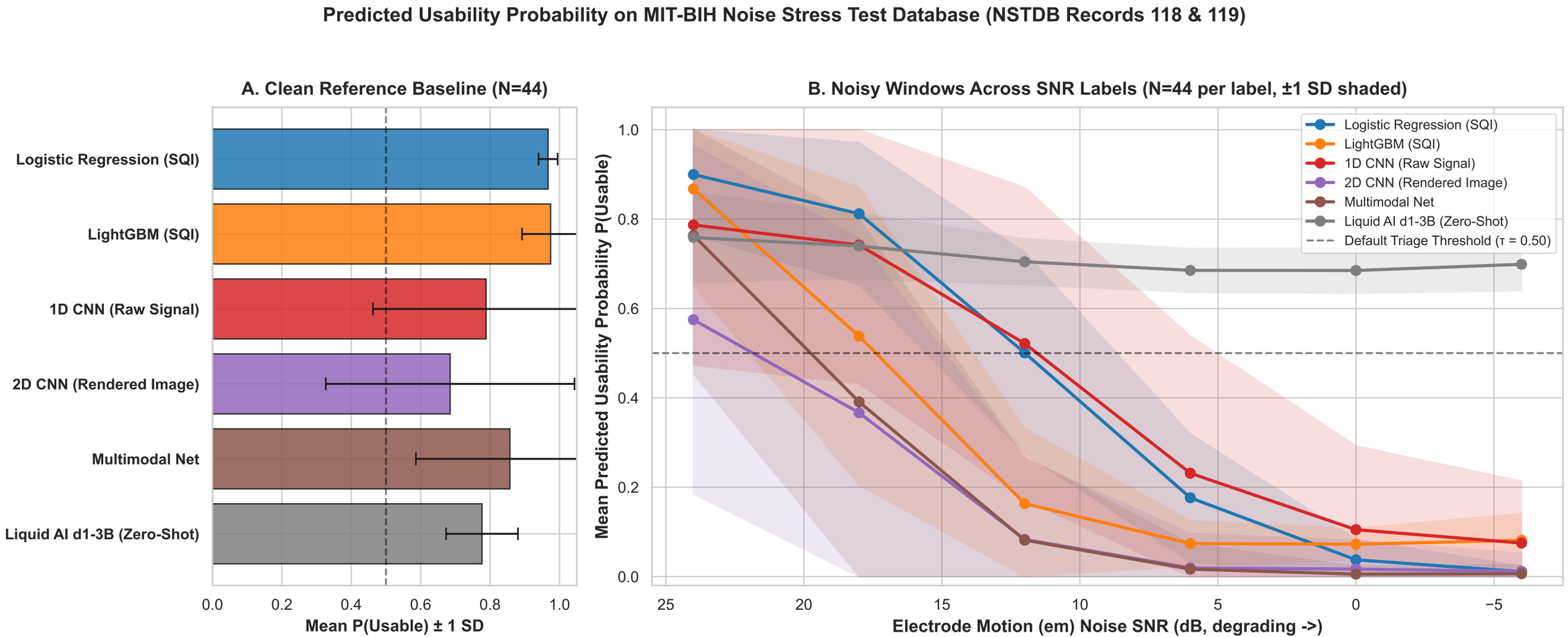
Predicted Usability Probability on MIT-BIH Noise Stress Test Database (NSTDB Records 118 & 119)
A. Clean Reference Baseline (N=44)
Logistic Regression (SQI)
LightGBM (SQI)
1D CNN (Raw Signal)
2D CNN (Rendered Image)
Multimodal Net
Liquid AI d1-3B (Zero-Shot)
0.0
0.2
0.4
0.6
0.8
1.0
Mean P(Usable) ± 1 SD
B. Noisy Windows Across SNR Labels (N=44 per label, ±1 SD shaded)
Mean Predicted Usability Probability P(Usable)
Logistic Regression (SQI)
LightGBM (SQI)
1D CNN (Raw Signal)
2D CNN (Rendered Image)
Multimodal Net
Liquid AI d1-3B (Zero-Shot)
Default Triage Threshold (τ = 0.50)
25
20
15
10
5
0
−5
Electrode Motion (em) Noise SNR (dB, degrading ->)